Wired Wear

Scientists in Germany have developed synthetic fibres that generate electricity when exposed to light. Researchers say the fibres could be woven into machine-washable clothes—the ultimate in portable solar cells. Just like the photovoltaic cells found in many pocket calculators, the new wires work by sandwiching three layers of malleable silicon between two conducting electrodes. One of the biggest challenges is making contact with each strand in a fabric. There are other issues too that need further work. Solar cells based on amorphous silicon are less efficient than their crystalline cousins. But they have advantages too. Besides being flexible, amorphous silicon is about a thousand times better at absorbing sunlight than the crystalline form. And the fashion conscious need not arch questioning brows as colour is not a problem. The fibre is transparent; it can be made to take on different colours by adjusting the thickness of the transparent protective coating. Depending upon the thickness of the layer, it could be made to look blue, brown or greenish.

Freeplay's self-powered radios and lanterns can be powered simply by winding each unit's handle

rely on gas or liquid, the aluminium fuel cell derives its power from a solid-state fuel. Because of the relative density of its power source, more energy can be liberated over a longer period of time at a considerably reduced cost. Experts estimate that the manufacturing costs of the aluminium fuel cell will be significantly lower than those of the lithium-ion or nickel metal hydride cells that are currently in use.

Traditional rechargeable batteries have a limited shelf life, low output capacity and require an auxiliary power source. The aluminium fuel cell can be recharged to its original capacity simply by replacing its rechargeable cartridge. This makes it ideal for use in instances where an auxiliary power source is unavailable.

Power dressing

In the not-so-distant future, you might literally get a charge out of hiking, walking or cycling. Your sweater could one day provide all the power you need to run your MP3 player, your mobile phone or your palmtop computer. If wearable computers are one of the recent revolutions in the digital industry, another is about using the kinetic movements of your body to charge electronic devices. And scientists are trying to figure out how exactly it can be done.

SRI International, a research firm in California, is working on boots that will convert the mechanical energy resulting from walking into electric power to charge gadgets, batteries and other devices. The idea for such an application came from an accidental discovery when SRI was working on artificial muscles. They created a polymer that contracted like a human muscle when electricity ran through it. They realised it could also work the other way round, creating electricity when it was compressed and released. To create electricity, the pressure could be applied while walking. The prototype boot generates about 0.5 watt of power, which is just about enough to recharge a cell phone. But Ron Pelrine, director of SRI, hopes that the boots' output could be raised to nearly 2 watts—enough power output to power a cell phone, a handheld computer, and a radio simultaneously. He estimates that the earliest we may see power-generating footwear hit the streets would be two years. And that depends on whether SRI receives the funding necessary to move the boots beyond the prototype stage. MIT Labs are also investigating heel generators, but the applications are not going to be dramatic given that power output will be severely limited.

This kind of promise has special applications